Assignment 1

Questions

Q1. Operator Precedence (0.5 marks)

Write a Python program that evaluates the following expression step by step:

$$(5 + 3) * 2 ** 2 - 10/2$$

Print the result and explain in comments how operator precedence affects the outcome.

Q2. Comparison and Logical Operators (0.5 marks)

Ask the user to input their age. Use comparison and logical operators to print whether the person is:

- A minor (under 18)
- An adult (18–64)
- A senior citizen (65 and above)

Q3. Conditional Flow – Temperature Advisor (0.5 marks)

Write a program that takes the current temperature as input and prints advice (e.g., "Freezing", "Chilly", "Warm", etc.) using if-elif-else.

Q4. Grade Classification (0.5 marks)

Prompt the user for a score (0–100). Classify it into letter grades (A, B, C, D, F). Include input validation to ensure the score is within range.

Q5. Shopping Discount Calculator (0.5 marks)

Ask the user for a purchase amount. Apply discounts based on thresholds (≥100: 20%, 50–99: 10%, <50: no discount). Print the final price.

Q6. Simple Authentication System (0.5 marks)

Simulate a login system. Predefine a username and password. Ask the user to input credentials and print whether access is granted or denied.

Q7. Conditional List Access Without Loops (0.5 marks)

Write a Python program that:

1. Creates a list of 5 integers: [12, 7, 25, 3, 18]
2. Uses individual if statements (not loops) to:
    - Check if the first item is greater than 10 and print it if true.
    - Check if the third item is even and print it if true.
    - Check if the last item is less than 20 and print it if true.
3. Create a new list called selected_values that includes only the values that passed the conditions above.
4. Print the selected_values list.

Q8. List Modification (0.5 marks)

Start with a list of 3 guests. Add one guest using append(), insert another at the beginning, and remove one using pop(). Print the final list.

Q9. Sorting and Error Handling (0.5 marks)

Create a list of 5 numbers entered by the user. Print the list in ascending and descending order. Then, attempt to access an out-of-range index and handle the error gracefully with a conditional check.

Q10. Integrated Problem – Movie Ticket System (0.5 marks)

Write a program that:

1. Asks the user for their age.
2. Determines ticket price based on age (under 5: free, 5–17: $5, 18–64: $10, 65+: $7).
3. Stores the ticket price in a list of daily sales.
4. Prints the updated list of sales.